Wasserstein Geometry and Generative Policies

Syllabus Notes

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Chapter 1

Probability Measures, Pushforwards, and Weak Calculus

1.1 Motivation and Scope

The goal of these notes is to introduce the minimal measure-theoretic and variational language required for modern formulations of:

- Wasserstein gradient flows,
- Schrödinger bridges,
- diffusion and drifting generative models,
- policy optimization in probability space.

Throughout these notes, probability distributions are treated as geometric objects that evolve in time. The central idea is that learning algorithms can often be interpreted not merely as optimization in parameter space, but as evolution equations over spaces of probability measures. This first phase introduces:

(i) probability measures on Euclidean spaces,

(ii) pushforward measures,

(iii) weak convergence,

(iv) entropy and Kullback–Leibler divergence,

(v) continuity equations and weak formulations.

We assume familiarity with basic probability theory, measure theory, and multivariable calculus.

1.2 Probability Measures on Euclidean Spaces

1.2.1 Borel probability measures

Let $\mathbb{R}^d$ denote Euclidean space equipped with its Borel σ-algebra $\mathcal{B}(\mathbb{R}^d)$.

Definition 1.1 (Probability measure)**.** A probability measure on $\mathbb{R}^d$ is a map

$$\mu : \mathcal{B}(\mathbb{R}^d) \to [0, 1]$$

such that:

(a) $\mu(\mathbb{R}^d) = 1$,

(b) for any countable collection of pairwise disjoint Borel sets $\{A_i\}_{i=1}^{\infty}$,

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i).$$

The set of all Borel probability measures on $\mathbb{R}^d$ is denoted by

$$\mathcal{P}(\mathbb{R}^d).$$

Definition 1.2 (Absolutely continuous measure)**.** A measure $\mu \in \mathcal{P}(\mathbb{R}^d)$ is absolutely continuous with respect to Lebesgue measure if there exists a nonnegative function

$$\rho \in L^1(\mathbb{R}^d)$$

with

$$\int_{\mathbb{R}^d} \rho(x)dx = 1$$

such that

$$\mu(A) = \int_A \rho(x)dx$$

for every Borel set A. The function ρ is called the density of μ.

In applied machine learning, probability measures are often represented implicitly through neural generators rather than explicit densities.

1.2.2 Moments

Definition 1.3 (Moment of order p)**.** For $p \geq 1$, the p-th moment of $\mu \in \mathcal{P}(\mathbb{R}^d)$ is

$$\int_{\mathbb{R}^d} \|x\|^p d\mu(x).$$

We define

$$\mathcal{P}_p(\mathbb{R}^d) = \left\{\mu \in \mathcal{P}(\mathbb{R}^d) : \int \|x\|^p d\mu(x) < \infty\right\}.$$

Finite second moments will become fundamental in Wasserstein geometry.

1.3 Random Variables and Pushforward Measures

1.3.1 Pushforwards

The pushforward operation is one of the most important constructions in modern generative modeling.

Definition 1.4 (Pushforward measure)**.** Let

$$T : \mathbb{R}^m \to \mathbb{R}^d$$

be measurable and let $\mu \in \mathcal{P}(\mathbb{R}^m)$. The pushforward of μ through T is the probability measure

$$T_{\#}\mu \in \mathcal{P}(\mathbb{R}^d)$$

defined by

$$(T_{\#}\mu)(A) = \mu(T^{-1}(A))$$

for every Borel set $A \subseteq \mathbb{R}^d$.

Remark 1.5. If

$$X \sim \mu$$

and

$$Y = T(X),$$

then

$$Y \sim T_{\#}\mu.$$

Thus pushforwards generalize the notion of the distribution of a transformed random variable.

1.3.2 Pushforwards in generative modeling

A generative model is frequently defined by:

(i) a latent distribution $z \sim \rho_0$,

(ii) a neural map

$$f_\theta : \mathbb{R}^m \to \mathbb{R}^d.$$

The generated distribution is

$$\rho_\theta = (f_\theta)_{\#}\rho_0.$$

This point of view is central in:

- normalizing flows,
- diffusion decoding maps,
- one-step generators,
- drifting models,
- policy distributions in reinforcement learning.

Example 1.6 (Policy as pushforward)**.** Suppose

$$z \sim \mathcal{N}(0, I)$$

and define a policy

$$a = f_\theta(s, z).$$

For fixed state s, the induced action distribution is

$$\pi_\theta(\cdot|s) = (f_\theta(s, \cdot))_{\#}\rho_0.$$

Thus a policy can be viewed as a state-dependent pushforward distribution.

1.3.3 Change of variables

Theorem 1.7 (Change-of-variables formula)**.** *Let*

$$T : \mathbb{R}^d \to \mathbb{R}^d$$

be a smooth bijection with smooth inverse. Suppose μ has density ρ. Then the pushforward measure

$$T_{\#}\mu$$

has density

$$\tilde{\rho}(y) = \rho(T^{-1}(y)) \left|\det DT^{-1}(y)\right|.$$

This theorem underlies likelihood computation in normalizing flows.

1.4 Weak Convergence of Measures

In optimization over distributions, one requires a notion of convergence between probability measures.

Definition 1.8 (Weak convergence)**.** A sequence

$$\mu_n \in \mathcal{P}(\mathbb{R}^d)$$

converges weakly to μ if

$$\int f \, d\mu_n \to \int f \, d\mu$$

for every bounded continuous function

$$f : \mathbb{R}^d \to \mathbb{R}.$$

We write

$$\mu_n \rightharpoonup \mu.$$

Remark 1.9. Weak convergence means convergence "in distribution."

Example 1.10. Let

$$\mu_n = \delta_{1/n}$$

where δ_x denotes the Dirac measure at x. Then

$$\mu_n \rightharpoonup \delta_0.$$

1.4.1 Tightness

Definition 1.11 (Tight family)**.** A family of measures

$$\mathcal{F} \subseteq \mathcal{P}(\mathbb{R}^d)$$

is tight if for every $\varepsilon > 0$ there exists a compact set $K \subseteq \mathbb{R}^d$ such that

$$\mu(K) \geq 1 - \varepsilon$$

for all $\mu \in \mathcal{F}$.

Theorem 1.12 (Prokhorov)**.** *A family of probability measures on $\mathbb{R}^d$ is relatively compact for weak convergence if and only if it is tight.*

This theorem is foundational in variational formulations over probability spaces.

1.5 Entropy and Kullback–Leibler Divergence

Entropy functionals play a central role in:

- diffusion models,
- Schrödinger bridges,
- entropy-regularized optimal transport,
- reinforcement learning.

1.5.1 Differential entropy

Definition 1.13 (Differential entropy)**.** Let μ have density ρ. The differential entropy is

$$\mathcal{H}(\rho) = -\int \rho(x) \log \rho(x) dx.$$

Entropy measures the spread or uncertainty of a distribution.

1.5.2 Relative entropy

Definition 1.14 (Kullback–Leibler divergence)**.** Let μ and ν be probability measures with densities ρ and η, respectively. The Kullback–Leibler divergence is

$$\mathrm{KL}(\mu \| \nu) = \int \rho(x) \log \frac{\rho(x)}{\eta(x)} dx$$

whenever μ is absolutely continuous with respect to ν.

Remark 1.15. The KL divergence is not symmetric:

$$\mathrm{KL}(\mu \| \nu) \neq \mathrm{KL}(\nu \| \mu).$$

Thus it is not a metric.

Theorem 1.16 (Gibbs inequality)**.** *For probability measures μ and ν,*

$$\mathrm{KL}(\mu \| \nu) \geq 0,$$

with equality if and only if

$$\mu = \nu$$

almost everywhere.

1.5.3 Forward and reverse KL

In machine learning, two optimization regimes appear frequently.

Forward KL

$$\mathrm{KL}(p \| q).$$

Penalizes missing support heavily.

Reverse KL

$$\mathrm{KL}(q \| p).$$

Often mode-seeking. Reverse KL objectives appear in policy optimization and drifting-field policies.

1.6 Curves of Measures and Continuity Equations

Modern optimal transport interprets probability distributions as evolving continuously in time.

1.6.1 Time-dependent measures

Consider a family of measures

$$(\rho_t)_{t\in[0,1]}.$$

Intuitively, mass moves through space according to a velocity field.

Definition 1.17 (Velocity field)**.** A velocity field is a map

$$v_t : \mathbb{R}^d \to \mathbb{R}^d.$$

The quantity

$$v_t(x)$$

represents the instantaneous velocity of mass at location x and time t.

1.6.2 Continuity equation

Definition 1.18 (Continuity equation)**.** The continuity equation is

$$\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0.$$

This equation expresses conservation of probability mass.

Remark 1.19. The continuity equation is analogous to conservation laws in fluid mechanics.

1.6.3 Weak formulation

Since densities may not be smooth, one frequently uses weak formulations.

Definition 1.20 (Weak solution of the continuity equation)**.** A curve of measures (ρ_t) is a weak solution of

$$\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0$$

if for every smooth compactly supported test function

$$\varphi \in C_c^\infty((0,1)\times\mathbb{R}^d),$$

one has

$$\int_0^1 \int_{\mathbb{R}^d} (\partial_t \varphi + \nabla\varphi \cdot v_t)\, d\rho_t dt = 0.$$

1.6.4 Particle interpretation

Suppose particles evolve according to the ODE

$$\dot{X}_t = v_t(X_t).$$

If

$$X_0 \sim \rho_0,$$

then the law of X_t evolves according to the continuity equation. Thus:

- particle dynamics induce measure dynamics,
- velocity fields induce trajectories in probability space.

This viewpoint becomes central in Wasserstein geometry.

1.7 Free Energies and Variational Perspectives

Many evolution equations can be written as gradient flows of an energy functional. A generic free energy has the form

$$\mathcal{F}(\mu) = \int V(x)\, d\mu(x) + \beta^{-1} \int \rho(x) \log \rho(x)\, dx, \qquad \mu = \rho(x)\, dx.$$

The first term represents potential energy with respect to the measure μ, while the second corresponds to entropy written in terms of its density ρ. Examples include:

- Gibbs measures,
- Langevin dynamics,
- diffusion models,
- entropy-regularized reinforcement learning.

In later phases we will show that many PDEs are gradient flows of such energies in Wasserstein space.

1.8 Summary and Outlook

The main ideas introduced in this phase are:

(1) Probability distributions can be viewed as geometric objects.

(2) Neural generators define pushforward measures.

(3) Learning can be interpreted as evolution in probability space.

(4) Entropy and KL divergences define variational objectives over measures.

(5) Continuity equations describe the transport of probability mass.

These concepts form the foundation for:

- Wasserstein distances,
- optimal transport,
- Wasserstein gradient flows,
- Schrödinger bridges,
- drifting generative models,
- generative policies in reinforcement learning.

In the next phase we introduce optimal transport and Wasserstein geometry.

Recommended Reading

1. C. Villani, *Topics in Optimal Transportation*

Chapter 2

Optimal Transport and Wasserstein Geometry

2.1 Motivation

Optimal transport (OT) studies the geometry of probability distributions by quantifying the cost of transporting mass from one distribution to another. The modern importance of OT in machine learning derives from several observations:

- probability distributions can be treated as geometric objects;
- Wasserstein distances induce meaningful topologies on spaces of measures;
- gradient flows in Wasserstein space produce important PDEs;
- generative models and policies evolve distributions during training;
- reinforcement learning objectives can often be interpreted as transport problems.

The goal of this phase is to introduce:

(i) Monge and Kantorovich optimal transport,

(ii) Wasserstein distances,

(iii) dynamic optimal transport,

(iv) Wasserstein geodesics,

(v) the geometry of W_2 space.

Throughout these notes we focus primarily on the quadratic Wasserstein metric W_2, since it is the relevant geometry for Wasserstein gradient flows and drifting-field policies.

2.2 The Monge Formulation

2.2.1 Mass transport maps

Suppose we are given two probability measures

$$\mu, \nu \in \mathcal{P}(\mathbb{R}^d).$$

Intuitively:

- μ represents an initial mass distribution,
- ν represents a target mass distribution.

The original formulation of optimal transport was introduced by Gaspard Monge in 1781.

Definition 2.1 (Transport map)**.** A measurable map

$$T : \mathbb{R}^d \to \mathbb{R}^d$$

is said to transport μ to ν if

$$T_{\#}\mu = \nu.$$

The goal is to find the transport map minimizing transportation cost.

Definition 2.2 (Monge problem)**.** Given a cost function

$$c : \mathbb{R}^d \times \mathbb{R}^d \to [0, \infty),$$

the Monge optimal transport problem is

$$\inf_T \int_{\mathbb{R}^d} c(x, T(x))d\mu(x)$$

subject to

$$T_{\#}\mu = \nu.$$

Typically one chooses

$$c(x, y) = \|x - y\|^p.$$

2.2.2 Difficulties of the Monge problem

The Monge formulation is highly nonlinear. Major difficulties include:

- existence of transport maps may fail;
- the pushforward constraint is nonlinear;
- transport maps cannot split mass.

Example 2.3. Suppose

$$\mu = \delta_0, \qquad \nu = \tfrac{1}{2}\delta_{-1} + \tfrac{1}{2}\delta_1.$$

No transport map can split the mass at 0 into two pieces.

These difficulties motivate the Kantorovich relaxation.

2.3 The Kantorovich Formulation

2.3.1 Transport plans

Instead of deterministic maps, Kantorovich introduced probabilistic couplings.

Definition 2.4 (Transport plan)**.** A transport plan between μ and ν is a probability measure

$$\gamma \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)$$

such that:

$$(\pi_x)_{\#}\gamma = \mu,$$
$$(\pi_y)_{\#}\gamma = \nu,$$

where

$$\pi_x(x, y) = x, \qquad \pi_y(x, y) = y.$$

The set of all such couplings is denoted

$$\Pi(\mu, \nu).$$

Remark 2.5. A transport plan specifies how much mass moves from x to y. Unlike Monge maps, transport plans may split mass.

2.3.2 Kantorovich optimal transport

Definition 2.6 (Kantorovich problem)**.** The Kantorovich optimal transport problem is

$$\inf_{\gamma\in\Pi(\mu,\nu)} \int_{\mathbb{R}^d\times\mathbb{R}^d} c(x, y)d\gamma(x, y).$$

This is a linear optimization problem over probability measures.

Theorem 2.7 (Existence of optimal plans)**.** *Let c be lower semicontinuous and bounded from below. Then for every*

$$\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$$

there exists at least one optimal transport plan.

2.4 Wasserstein Distances

2.4.1 Definition

The most important costs are powers of Euclidean distance.

Definition 2.8 (p-Wasserstein distance)**.** Let $p \geq 1$. For

$$\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d),$$

the p-Wasserstein distance is

$$W_p(\mu, \nu) = \left(\inf_{\gamma\in\Pi(\mu,\nu)} \int \|x - y\|^p d\gamma(x, y) \right)^{1/p}.$$

The most important case for modern machine learning is:

$$W_2.$$

2.4.2 Metric structure

Theorem 2.9. *For $p \geq 1$, the Wasserstein distance defines a metric on*

$$\mathcal{P}_p(\mathbb{R}^d).$$

Thus probability distributions become points in a metric space.

Remark 2.10. Unlike KL divergence, Wasserstein distances compare distributions geometrically. Even distributions with disjoint supports may have finite Wasserstein distance.

2.4.3 Comparison with KL divergence

Suppose

$$\mu = \delta_0, \qquad \nu = \delta_1.$$

Then:

$$W_2(\mu, \nu) = 1,$$

while

$$\mathrm{KL}(\mu \| \nu) = \infty.$$

This sensitivity to geometry is one reason Wasserstein distances are important in generative modeling.

2.5 Optimal Maps and Brenier Theory

The quadratic Wasserstein geometry possesses remarkable structure.

2.5.1 Brenier theorem

Theorem 2.11 (Brenier). *Let*

$$\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$$

and assume μ is absolutely continuous. Then there exists a unique optimal transport map

$$T : \mathbb{R}^d \to \mathbb{R}^d$$

for the quadratic cost

$$c(x, y) = \|x - y\|^2.$$

Moreover,

$$T = \nabla \varphi$$

for a convex function

$$\varphi : \mathbb{R}^d \to \mathbb{R}.$$

Remark 2.12. This theorem is foundational for Wasserstein geometry. Optimal transport maps are gradients of convex potentials.

2.5.2 Displacement interpolation

Given the optimal map T, define

$$T_t(x) = (1 - t)x + tT(x).$$

Then the interpolating measures

$$\mu_t = (T_t)_{\#}\mu$$

define the Wasserstein geodesic between μ and ν. This interpolation is called displacement interpolation.

2.6 Dynamic Optimal Transport

The static Kantorovich formulation can be rewritten dynamically. This viewpoint is one of the most important developments in modern OT.

2.6.1 Continuity equations

Recall the continuity equation:

$$\partial_t \rho_t + \mathrm{div}(\rho_t v_t) = 0.$$

Here:

- ρ_t is a time-dependent density,
- v_t is a velocity field.

The velocity field transports mass continuously.

2.6.2 Benamou–Brenier formulation

Theorem 2.13 (Benamou–Brenier)**.** *Let*

$$\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R}^d).$$

Then

$$W_2^2(\mu_0, \mu_1) = \inf_{(\rho_t, v_t)} \int_0^1 \int_{\mathbb{R}^d} \|v_t(x)\|^2 \rho_t(x)\, dx dt$$

subject to:

$$\begin{aligned} \partial_t \rho_t + \mathrm{div}(\rho_t v_t) = 0, \\ \rho_0 = \mu_0, \\ \rho_1 = \mu_1. \end{aligned}$$

A graduate-note proof of the Benamou–Brenier formula is included separately.

2.6.3 Interpretation

The quantity

$$\int \|v_t(x)\|^2 \rho_t(x) dx$$

represents kinetic energy. Thus:

> The Wasserstein distance is the minimum kinetic energy required to transport one distribution into another.

This dynamic viewpoint is central for:

- Wasserstein gradient flows,
- fluid formulations of generative models,
- Schrödinger bridges,
- policy transport in RL.

2.7 Wasserstein Geodesics

2.7.1 Geodesic curves

A geodesic is a shortest path in a metric space.

Definition 2.14 (Geodesic in metric space)**.** Let (X, d) be a metric space. A curve

$$\gamma_t : [0, 1] \to X$$

is a geodesic if

$$d(\gamma_s, \gamma_t) = |t - s| d(\gamma_0, \gamma_1)$$

for all $s, t \in [0, 1]$.

2.7.2 Wasserstein geodesics

The displacement interpolation

$$\mu_t = (T_t)_{\#}\mu$$

constructed from the Brenier map defines a geodesic in $(\mathcal{P}_2(\mathbb{R}^d), W_2)$. Thus:

> Optimal transport equips probability space with a Riemannian-like geometry.

2.7.3 Velocity representation

Along Wasserstein geodesics, the velocity field satisfies:

$$\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0.$$

The velocity field minimizing kinetic energy defines the geodesic flow.

2.8 Otto Calculus and Formal Riemannian Structure

Felix Otto observed that W_2 space behaves formally like an infinite-dimensional Riemannian manifold.

2.8.1 Tangent vectors

Suppose ρ_t evolves according to

$$\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0.$$

The tangent vector at ρ can formally be identified with

$$\dot{\rho} = -\operatorname{div}(\rho v).$$

Among all velocity fields generating the same tangent vector, the minimal kinetic-energy representative is gradient-valued:

$$v = \nabla \phi.$$

Thus tangent vectors may be represented as:

$$\dot{\rho} = -\operatorname{div}(\rho \nabla \phi).$$

2.8.2 Formal metric tensor

Otto's formal metric is:

$$\langle \dot{\rho}_1, \dot{\rho}_2 \rangle_\rho = \int_{\mathbb{R}^d} \nabla \phi_1 \cdot \nabla \phi_2 \, d\rho.$$

This provides the formal Riemannian structure underlying Wasserstein gradient flows.

Remark 2.15. The space $\mathcal{P}_2(\mathbb{R}^d)$ is not literally a finite-dimensional manifold. The Otto calculus is partly formal, but extraordinarily powerful.

2.9 Convexity Along Geodesics

In Euclidean optimization, convexity governs gradient descent. In Wasserstein geometry, one instead studies convexity along geodesics.

Definition 2.16 (Displacement convexity)**.** A functional

$$\mathcal{F} : \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$$

is displacement convex if

$$\mathcal{F}(\mu_t)$$

is convex along Wasserstein geodesics.

Examples include:

- entropy functionals,
- internal energies,
- interaction energies.

Displacement convexity is fundamental for:

- uniqueness of gradient flows,
- convergence of diffusion equations,
- entropy dissipation.

2.10 Wasserstein Geometry in Machine Learning

2.10.1 Generative modeling

Generative models induce distributions

$$\rho_\theta = (f_\theta)_\# \rho_0.$$

Training changes the generated distribution through parameter updates. The induced trajectory:

$$\theta_t \mapsto \rho_{\theta_t}$$

may sometimes be interpreted as an approximate Wasserstein gradient flow.

2.10.2 Diffusion and flow models

Diffusion models define stochastic transport processes between distributions. Flow matching and continuous normalizing flows define velocity fields transporting probability mass. Schrödinger bridges regularize optimal transport with entropy.

2.10.3 Policies as evolving distributions

In reinforcement learning:

$$\pi(a|s)$$

is a conditional probability distribution over actions. Modern policy optimization methods may be interpreted as transport in probability space:

$$\pi_t \to \pi_{t+1}.$$

This perspective motivates Wasserstein formulations of RL and drifting-field policies.

2.11 Summary

The key ideas of this phase are:

(1) Optimal transport studies the geometry of probability distributions.

(2) The Monge formulation searches for deterministic transport maps.

(3) The Kantorovich relaxation allows probabilistic couplings.

(4) Wasserstein distances metrize spaces of probability measures.

(5) The quadratic Wasserstein metric admits a dynamic fluid formulation.

(6) Wasserstein geodesics describe optimal displacement of mass.

(7) The Otto calculus provides a formal Riemannian structure on probability space.

(8) Many learning algorithms may be interpreted as evolution equations in Wasserstein space.

The central conceptual shift is the following:

> Instead of optimizing points in Euclidean space, one optimizes probability distributions in Wasserstein space.

This viewpoint underlies:

- Wasserstein gradient flows,
- diffusion and flow models,
- Schrödinger bridges,
- entropy-regularized optimal transport,
- modern formulations of reinforcement learning.

In the next phase we introduce Wasserstein gradient flows and their connections with PDEs, variational principles, and learning dynamics.

Recommended Reading

1. C. Villani, *Optimal Transport: Old and New.*

2. G. Peyré and M. Cuturi, *Computational Optimal Transport.*

3. F. Santambrogio, *Optimal Transport for Applied Mathematicians.*

4. J.-D. Benamou and Y. Brenier, "A Computational Fluid Mechanics Solution to the Monge–Kantorovich Mass Transfer Problem."

5. F. Otto, "The Geometry of Dissipative Evolution Equations."

Chapter 3

Wasserstein Gradient Flows

3.1 Motivation

In Euclidean optimization, gradient descent evolves points according to:

$$\dot{x}_t = -\nabla f(x_t).$$

The fundamental idea of Wasserstein gradient flows is the following:

> Instead of evolving points in Euclidean space, evolve probability distributions in Wasserstein space.

The resulting framework connects:

- partial differential equations,
- diffusion processes,
- entropy dissipation,
- stochastic control,
- generative modeling,
- reinforcement learning.

This chapter develops the formalism of Wasserstein gradient flows (WGF), emphasizing:

(i) variational principles,

(ii) continuity equations,

(iii) free energy functionals,

(iv) the Jordan–Kinderlehrer–Otto (JKO) scheme,

(v) connections to diffusion and learning.

3.2 Gradient Flows in Euclidean Space

We briefly recall ordinary gradient descent.

3.2.1 Gradient descent ODE

Let

$$f : \mathbb{R}^d \to \mathbb{R}$$

be smooth.

The gradient flow equation is

$$\dot{x}_t = -\nabla f(x_t).$$

This equation follows the direction of steepest descent.

3.2.2 Energy dissipation

Along solutions:

$$\frac{d}{dt} f(x_t) = \nabla f(x_t) \cdot \dot{x}_t.$$

Substituting the gradient flow equation gives:

$$\frac{d}{dt} f(x_t) = -\|\nabla f(x_t)\|^2 \leq 0.$$

Thus gradient flows dissipate energy monotonically.

3.2.3 Metric interpretation

Gradient descent depends on the geometry of the underlying space.

In Euclidean space, the metric tensor is the identity.

On a finite-dimensional Riemannian manifold, gradient descent is defined by using the metric to convert the differential of an objective into a vector field.

In Wasserstein geometry, the metric structure is different, leading to different gradient flows.

3.3 Functionals on Probability Space

Wasserstein gradient flows evolve probability measures to decrease an energy functional.

3.3.1 Energy functionals

A functional is a map:

$$\mathcal{F} : \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}.$$

Important examples include:

(a) potential energies,

(b) entropy functionals,

(c) interaction energies,

(d) KL divergences.

3.3.2 Potential energy

Given a confining potential

$$V : \mathbb{R}^d \to \mathbb{R},$$

define:

$$\mathcal{V}(\rho) = \int V(x) d\rho(x).$$

3.3.3 Entropy

For a density ρ:

$$\mathcal{H}(\rho) = \int \rho(x) \log \rho(x) dx.$$

Note the sign convention differs from Shannon entropy.

3.3.4 Free energy

A central object is the free energy:

$$\mathcal{F}(\rho) = \int V(x) d\rho(x) + \beta^{-1} \int \rho(x) \log \rho(x) dx.$$

This combines:

- energy minimization,
- entropy regularization.

The parameter:

$$\beta^{-1}$$

plays the role of temperature.

3.4 First Variations

To define gradient flows on probability space we require a notion of derivative.

3.4.1 Functional derivatives

Definition 3.1 (First variation)**.** Let $\mathcal{F}$ be a functional defined on a class of probability densities. The first variation of $\mathcal{F}$ at ρ is a function

$$\frac{\delta \mathcal{F}}{\delta \rho}(\rho)$$

such that, for every smooth perturbation η satisfying

$$\int_{\mathbb{R}^d} \eta(x)\, dx = 0,$$

one has

$$\frac{d}{d\varepsilon} \mathcal{F}(\rho + \varepsilon \eta) \Big|_{\varepsilon=0} = \int_{\mathbb{R}^d} \frac{\delta \mathcal{F}}{\delta \rho}(\rho)(x)\, \eta(x)\, dx.$$

The zero-mass condition on η ensures that $\rho + \varepsilon\eta$ remains a probability density to first order. On probability densities, the first variation is defined only up to an additive constant, since constants vanish against zero-mass perturbations.

3.4.2 Examples

Potential energy For

$$\mathcal{V}(\rho) = \int V \, d\rho,$$

one has:

$$\frac{\delta \mathcal{V}}{\delta \rho} = V.$$

Entropy For

$$\mathcal{H}(\rho) = \int \rho \log \rho,$$

one obtains:

$$\frac{\delta \mathcal{H}}{\delta \rho} = \log \rho + 1.$$

A proof of this first-variation computation is included separately.

Free energy For

$$\mathcal{F}(\rho) = \int V \, d\rho + \beta^{-1} \int \rho \log \rho,$$

we get:

$$\frac{\delta \mathcal{F}}{\delta \rho} = V + \beta^{-1}(\log \rho + 1).$$

3.5 Wasserstein Gradient Flows

3.5.1 Formal definition

The Wasserstein gradient flow of a functional $\mathcal{F}$ is formally:

$$\partial_t \rho_t = \operatorname{div}\left(\rho_t \nabla \frac{\delta \mathcal{F}}{\delta \rho}(\rho_t)\right).$$

Equivalently:

$$\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0,$$

with velocity field:

$$v_t = -\nabla \frac{\delta \mathcal{F}}{\delta \rho}(\rho_t).$$

3.5.2 Interpretation

Compare with Euclidean gradient flow:

$$\dot{x}_t = -\nabla f(x_t).$$

In Wasserstein space:

- points become distributions,
- velocities become transport vector fields,
- gradients become functional derivatives.

3.6 The Fokker–Planck Equation

One of the most important examples of a Wasserstein gradient flow is the Fokker–Planck equation.

3.6.1 Langevin dynamics

Consider the stochastic differential equation:

$$dX_t = -\nabla V(X_t)dt + \sqrt{2\beta^{-1}}dW_t.$$

Its probability density evolves according to:

$$\partial_t \rho_t = \operatorname{div}(\rho_t \nabla V) + \beta^{-1}\Delta \rho_t.$$

3.6.2 Gradient flow structure

Theorem 3.2. *The Fokker–Planck equation is the Wasserstein gradient flow of the free energy:*

$$\mathcal{F}(\rho) = \int V d\rho + \beta^{-1}\int \rho \log \rho.$$

3.6.3 Derivation

Using:

$$\frac{\delta \mathcal{F}}{\delta \rho} = V + \beta^{-1}(\log \rho + 1),$$

we compute:

$$\begin{aligned}\partial_t \rho &= \operatorname{div}\left(\rho \nabla (V + \beta^{-1}\log \rho)\right) \\ &= \operatorname{div}(\rho \nabla V) + \beta^{-1}\operatorname{div}(\rho \nabla \log \rho).\end{aligned}$$

Since

$$\rho \nabla \log \rho = \nabla \rho,$$

we obtain:

$$\partial_t \rho = \operatorname{div}(\rho \nabla V) + \beta^{-1}\Delta \rho.$$

3.7 Energy Dissipation

Gradient flows dissipate free energy.

Theorem 3.3 (Energy dissipation)**.** *Along Wasserstein gradient flows:*

$$\frac{d}{dt}\mathcal{F}(\rho_t) = -\int \left\|\nabla \frac{\delta \mathcal{F}}{\delta \rho}\right\|^2 d\rho_t.$$

Thus:

$$\frac{d}{dt}\mathcal{F}(\rho_t) \leq 0.$$

Remark 3.4. This is the infinite-dimensional analogue of:

$$\frac{d}{dt}f(x_t) = -\|\nabla f(x_t)\|^2.$$

3.8 The Jordan–Kinderlehrer–Otto Scheme

The JKO scheme gives a variational discretization of Wasserstein gradient flows.

3.8.1 Implicit Euler in Wasserstein space

Recall ordinary gradient descent:

$$x_{k+1} = \underset{x}{\operatorname{argmin}} \left(\frac{1}{2\tau} \|x - x_k\|^2 + f(x) \right).$$

The JKO scheme replaces Euclidean distance with Wasserstein distance.

Definition 3.5 (JKO scheme)**.** Given timestep $\tau > 0$, define recursively:

$$\rho_{k+1} = \underset{\rho}{\operatorname{argmin}} \left(\frac{1}{2\tau} W_2^2(\rho, \rho_k) + \mathcal{F}(\rho) \right).$$

3.8.2 Interpretation

Each step balances:

- staying close to the previous distribution,
- decreasing free energy.

Thus Wasserstein gradient flows emerge from repeated proximal optimization.

3.8.3 Conceptual importance

The JKO scheme is foundational because it:

- defines gradient flows variationally,
- avoids explicit PDE analysis,
- links optimization and transport geometry.

Many modern ML methods can be interpreted as approximate JKO steps.

3.9 KL Divergence as Free Energy

Suppose:

$$\pi(x) \propto e^{-V(x)}.$$

Then:

$$\mathrm{KL}(\rho \| \pi) = \int \rho \log \rho + \int V \, d\rho + \text{constant}.$$

Thus:

KL divergence is a free energy.

Consequently:

> The Fokker–Planck equation is gradient descent on KL divergence in Wasserstein space.

This observation is central for:

- diffusion models,
- score-based modeling,
- entropy-regularized RL,
- Schrödinger bridges.

3.10 Connections to Generative Modeling

3.10.1 Diffusion models

Diffusion models gradually transform noise distributions into data distributions.

The forward diffusion process satisfies a Fokker–Planck equation.

The reverse process can be interpreted as transport in probability space.

3.10.2 Flow models

Continuous normalizing flows evolve densities according to:

$$\partial_t \rho_t + \mathrm{div}(\rho_t v_t) = 0.$$

Thus they directly implement deterministic transport.

3.10.3 Drifting models

Drifting models train generators whose induced distributions evolve during optimization.

A key question is:

> Does SGD in parameter space induce approximate Wasserstein gradient flow in distribution space?

This perspective motivates drifting generative modeling.

3.11 Connections to Reinforcement Learning

Policies are conditional distributions:

$$\pi(a|s).$$

Policy optimization may be interpreted as transport in action-distribution space.

Instead of:

$$\theta_t \to \theta_{t+1},$$

one may study:

$$\pi_t \to \pi_{t+1}.$$

Recent work formulates RL as Wasserstein gradient flow toward soft optimal policies.

3.11.1 Entropy-regularized RL

Soft RL objectives often involve:

$$\mathbb{E}[-Q(a, s)] + \beta^{-1}\mathrm{KL}(\pi\|\pi_0).$$

This resembles free-energy minimization.
Consequently:

> Soft policy optimization may be viewed as gradient flow in probability space.

This viewpoint underlies drifting-field policies.

3.12 Otto Calculus Revisited

Recall the formal metric:

$$\langle \dot{\rho}_1, \dot{\rho}_2 \rangle_\rho = \int \nabla\phi_1 \cdot \nabla\phi_2 d\rho.$$

Under this geometry:

$$\partial_t \rho = -\mathrm{grad}_{W_2}\mathcal{F}(\rho).$$

Thus Wasserstein gradient flow is literally gradient descent under the Wasserstein metric.
This is the conceptual core of the theory.

3.13 Summary

The main ideas of this phase are:

(1) Wasserstein space admits gradient-flow dynamics.

(2) Functionals on probability space define evolution equations.

(3) The Fokker–Planck equation is a Wasserstein gradient flow.

(4) Free-energy minimization combines energy and entropy.

(5) The JKO scheme defines variational time discretization.

(6) KL divergence can be interpreted as a free energy.

(7) Many modern AI models evolve distributions rather than points.

The key conceptual shift is:

> Optimization algorithms may be interpreted as transport dynamics in probability space.

This idea will be fundamental for:

- Schrödinger bridges,
- entropy-regularized optimal transport,
- diffusion models,

- score-based generative modeling,
- Wasserstein reinforcement learning,
- drifting-field policies.

In the next phase we study policy optimization and reinforcement learning directly in Wasserstein space.

Recommended Reading

1. L. Ambrosio, N. Gigli, and G. Savaré, *Gradient Flows in Metric Spaces and in the Space of Probability Measures.*
2. F. Otto, "The Geometry of Dissipative Evolution Equations."
3. R. Jordan, D. Kinderlehrer, and F. Otto, "The Variational Formulation of the Fokker–Planck Equation."
4. C. Villani, *Optimal Transport: Old and New.*
5. F. Santambrogio, *Optimal Transport for Applied Mathematicians.*

Chapter 4

Entropy, Free Energy, Gibbs Measures, and Variational Principles

4.1 Motivation

The previous chapter showed that many evolution equations may be interpreted as gradient flows in Wasserstein space. The next step is to understand the functionals that generate these flows.

Three objects appear repeatedly throughout modern machine learning:

1. Gibbs distributions,
2. Kullback–Leibler divergence,
3. entropy-regularized optimization.

These concepts form the common mathematical foundation of:

- diffusion models,
- Schrödinger bridges,
- variational inference,
- soft reinforcement learning,
- Wasserstein policy optimization,
- drifting-field policies.

The main goal of this chapter is to understand the variational principle

$$\pi = \operatorname*{argmin}_{\rho} \left(\int V \, d\rho + \beta^{-1} \int \rho \log \rho \right),$$

which lies at the heart of all subsequent developments.

4.2 Entropy

4.2.1 Differential entropy

Let ρ be a probability density on $\mathbb{R}^d$.

Definition 4.1 (Differential entropy)**.** The differential entropy of ρ is

$$H(\rho) = -\int_{\mathbb{R}^d} \rho(x) \log \rho(x)\, dx.$$

Entropy measures uncertainty or spread.

Example 4.2. Among all distributions with fixed variance, the Gaussian maximizes entropy.

Remark 4.3. In mathematics it is often convenient to use

$$\mathcal{H}(\rho) = \int \rho \log \rho,$$

which differs by a sign from Shannon entropy.

4.2.2 Relative entropy

Definition 4.4 (Kullback–Leibler divergence)**.** Let ρ and π be probability densities.
The relative entropy is

$$\mathrm{KL}(\rho \| \pi) = \int \rho(x) \log \frac{\rho(x)}{\pi(x)}\, dx.$$

Theorem 4.5 (Gibbs inequality)**.**

$$\mathrm{KL}(\rho \| \pi) \geq 0.$$

Equality holds iff

$$\rho = \pi$$

almost everywhere.

Proof. Since

$$\log u \leq u - 1,$$

one obtains

$$-\log u \geq 1 - u.$$

Substituting

$$u = \frac{\pi(x)}{\rho(x)}$$

and integrating gives the claim.

□

4.3 Gibbs Measures

4.3.1 Definition

Let

$$V : \mathbb{R}^d \to \mathbb{R}$$

be a potential energy.

Definition 4.6 (Partition function)**.**

$$Z = \int_{\mathbb{R}^d} e^{-\beta V(x)} dx.$$

Definition 4.7 (Gibbs measure)**.** The Gibbs density associated with V is

$$\pi(x) = \frac{1}{Z} e^{-\beta V(x)}.$$

Remark 4.8. The parameter

$$\beta^{-1}$$

is interpreted as temperature.

4.3.2 Examples

Quadratic potential

$$V(x) = \frac{1}{2}\|x\|^2.$$

Then

$$\pi(x) \propto e^{-\beta\|x\|^2/2},$$

which is Gaussian.

Softmax distributions If the state space is finite,

$$V_i = -Q_i,$$

then

$$\pi_i = \frac{e^{Q_i/\alpha}}{\sum_j e^{Q_j/\alpha}}.$$

This is precisely the Boltzmann policy appearing in soft RL.

4.4 Free Energy

4.4.1 Definition

Definition 4.9 (Free energy)**.** The free energy functional is

$$\mathcal{F}(\rho) = \int V(x)\rho(x)\, dx + \beta^{-1} \int \rho(x) \log \rho(x)\, dx.$$

The first term is expected energy.
The second term penalizes low entropy.

4.4.2 Physical interpretation

The classical thermodynamic identity is

$$F = E - TS.$$

In our notation,

$$\mathcal{F} = \text{energy} + \beta^{-1}\text{entropy}.$$

Because entropy carries a minus sign, this is exactly the same formula.

4.5 The Gibbs Variational Principle

We now arrive at one of the central theorems of the entire syllabus.

Theorem 4.10 (Gibbs variational principle)**.** *Let*

$$\pi(x) = \frac{1}{Z} e^{-\beta V(x)}.$$

Then

$$\pi = \operatorname*{argmin}_{\rho} \mathcal{F}(\rho).$$

Proof. Observe that

$$\log \pi(x) = -\beta V(x) - \log Z.$$

Therefore

$$V(x) = -\beta^{-1} \log \pi(x) - \beta^{-1} \log Z.$$

Substituting into the free energy gives

$$\begin{aligned} \mathcal{F}(\rho) &= \int \left(-\beta^{-1} \log \pi - \beta^{-1} \log Z\right) \rho + \beta^{-1} \int \rho \log \rho \\ &= \beta^{-1} \int \rho \log \frac{\rho}{\pi} - \beta^{-1} \log Z. \end{aligned}$$

Hence

$$\mathcal{F}(\rho) = \beta^{-1} \mathrm{KL}(\rho \| \pi) - \beta^{-1} \log Z.$$

Since KL divergence is nonnegative,

$$\mathcal{F}(\rho) \geq -\beta^{-1} \log Z.$$

Equality occurs iff

$$\rho = \pi.$$

□

Chapter 5

Schrödinger Bridges and Non-Conservative Transport

5.1 Motivation

The previous chapter introduced entropy, Gibbs measures, and variational principles. We now use those ideas to move from deterministic optimal transport to stochastic transport.

The central object of this chapter is the *Schrödinger bridge*. At a high level, a Schrödinger bridge asks for the most likely stochastic evolution connecting two prescribed probability distributions. In modern generative modeling language, it asks for a probability path

$$\rho_0 = \rho_a, \qquad \rho_1 = \rho_b,$$

together with a velocity or drift field that transports samples from the initial marginal to the terminal marginal.

The new feature is that the path is not chosen only by shortest transport cost. It is regularized by diffusion, entropy, or a reference stochastic process. This makes Schrödinger bridges a natural meeting point of:

- optimal transport,
- stochastic control,
- diffusion models,
- multi-marginal interpolation,
- policy improvement under entropy regularization.

For our long-term goal, the important question is slightly sharper:

What happens if the bridge is not energy-conserving?

This question is the doorway to contact geometry.

5.2 From Optimal Transport to Schrödinger Bridges

5.2.1 The deterministic transport picture

In the Benamou–Brenier formulation, the squared Wasserstein distance is obtained by minimizing kinetic energy:

$$W_2^2(\rho_a, \rho_b) = \inf_{\rho_t, v_t} \int_0^1 \int_{\mathbb{R}^d} \|v_t(x)\|^2 \rho_t(x)\, dx\, dt,$$

subject to

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0, \qquad \rho_0 = \rho_a, \qquad \rho_1 = \rho_b.$$

The continuity equation says that probability mass is transported by the velocity field v_t. There is no creation or destruction of mass.

5.2.2 Adding diffusion

A Schrödinger bridge replaces the deterministic continuity equation by a diffusive equation of Fokker–Planck type:

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = \varepsilon \Delta \rho_t.$$

The diffusion coefficient $\varepsilon > 0$ measures the strength of the stochastic perturbation. The velocity v_t now controls a noisy transport process rather than a purely deterministic one.

Remark 5.1. When $\varepsilon \to 0$, the Schrödinger bridge problem formally approaches optimal transport. For positive ε, the bridge prefers stochastic paths that are close to a reference diffusion.

5.3 The Generalized Schrödinger Bridge

The version that will matter for us includes a potential energy U. The action is

$$J(\rho, v) = \int_0^1 \int_{\mathbb{R}^d} \left(\frac{1}{2} \|v_t(x)\|^2 + U(x) \right) \rho_t(x)\, dx\, dt.$$

The first term is kinetic energy. The second term penalizes or rewards regions of state space through the potential U.

Definition 5.2 (Generalized Schrödinger bridge)**.** A generalized Schrödinger bridge between ρ_a and ρ_b is a solution of

$$\inf_{\rho_t, v_t} \int_0^1 \int_{\mathbb{R}^d} \left(\frac{1}{2} \|v_t(x)\|^2 + U(x) \right) \rho_t(x)\, dx\, dt$$

subject to

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = \varepsilon \Delta \rho_t, \qquad \rho_0 = \rho_a, \qquad \rho_1 = \rho_b.$$

One may also impose intermediate marginal constraints

$$\rho_{t_m} = \rho_m, \qquad m = 1, \ldots, M.$$

This is important for applications where one observes snapshots of a system at several times but not the full underlying trajectory.

5.4 Hamiltonian Structure

The optimality conditions of the generalized bridge can be written in Hamiltonian form. Introduce a scalar field $S_t(x)$, which plays the role of a momentum or cotangent variable. At the optimum, the velocity is a gradient:

$$v_t(x) = \nabla S_t(x).$$

The density and momentum then evolve according to a coupled system of the form

$$\partial_t \rho_t = -\nabla \cdot (\rho_t \nabla S_t),$$

and

$$\partial_t S_t = -\frac{1}{2}\|\nabla S_t\|^2 + \frac{1}{2}\varepsilon^2 \frac{\delta I}{\delta \rho}(\rho_t) + U.$$

Here

$$I(\rho) = \int_{\mathbb{R}^d} \|\nabla \log \rho(x)\|^2 \rho(x)\, dx$$

is the Fisher information.

Remark 5.3. The field S_t should be read as the cotangent representative of the motion. The density derivative $\partial_t \rho_t$ is tangent to the probability manifold, while S_t is the dual object whose gradient gives the velocity field moving the density.

The corresponding Hamiltonian is naturally split into kinetic and potential parts:

$$H(\rho, S) = \underbrace{\frac{1}{2}\int \|\nabla S(x)\|^2 \rho(x)\, dx}_{K(\rho,S)} + \underbrace{F(\rho)}_{\text{potential contribution}}.$$

The Fisher information belongs to the potential contribution. It is the trace left by stochastic diffusion after rewriting the bridge in Hamiltonian variables.

5.5 The Jacobi Metric Viewpoint

Classical mechanics has a useful geometric reformulation: under suitable conditions, Hamiltonian trajectories can be seen as geodesics for a rescaled metric called a Jacobi metric.

In the Wasserstein setting, the base metric is the Wasserstein metric g_{W_2}. The generalized bridge can be viewed as a geodesic problem for a conformal rescaling

$$g_J = \big(H - F(\rho)\big) g_{W_2}.$$

The factor $H - F(\rho)$ measures the kinetic part of the energy. Thus the potential does not replace the Wasserstein metric; it changes the geometry by making some regions of probability space more expensive than others.

Remark 5.4. This is the first point where the bridge stops looking like ordinary optimal transport. The geometry is no longer only the bare Wasserstein geometry of mass movement. It is Wasserstein geometry shaped by an energy landscape.

5.6 Why Conservative Bridges Are Not Enough

The Hamiltonian formulation above is conservative: the total Hamiltonian is constant along the idealized flow. This is appropriate for systems whose energy is preserved, but many systems of interest in generative modeling and reinforcement learning are not like that.

Examples include:

- a policy distribution changing as new value information is injected,
- a generative model being guided toward a condition,
- a biological population differentiating over time,
- an offline policy being updated toward an online-improved policy.

In these cases, the path may gain or lose effective energy. The target itself may drift. The geometry should therefore allow the bridge to be non-conservative.

5.7 The Non-Conservative Lift

The non-conservative bridge introduces one extra scalar variable z_t. This variable stores the accumulated action along the path. Instead of minimizing a fixed action directly, one prescribes its evolution:

$$\partial_t z_t = \int_{\mathbb{R}^d} \left(\frac{1}{2} \|v_t(x)\|^2 + U(x) \right) \rho_t(x)\, dx - \gamma z_t.$$

The parameter $\gamma \in \mathbb{R}$ controls damping or amplification. The bridge is still constrained by the Fokker–Planck equation

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = \varepsilon \Delta \rho_t.$$

Definition 5.5 (Non-conservative generalized Schrödinger bridge)**.** A non-conservative generalized Schrödinger bridge is a controlled density path (ρ_t, v_t, z_t) satisfying

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = \varepsilon \Delta \rho_t$$

and

$$\partial_t z_t = \int_{\mathbb{R}^d} \left(\frac{1}{2} \|v_t(x)\|^2 + U(x) \right) \rho_t(x)\, dx - \gamma z_t,$$

with prescribed endpoint marginals, and possibly prescribed intermediate marginals.

The point of z_t is not that probability mass is no longer conserved. The mass is still conserved by the Fokker–Planck equation. What is no longer conserved is the energy/action budget governing the bridge.

5.8 Contact Hamiltonian Dynamics

Contact geometry is the odd-dimensional analogue of symplectic geometry. The usual Hamiltonian phase space is enlarged by one scalar coordinate. In our case, the enlarged phase space is formally

$$T^*\mathcal{P}^+(\mathbb{R}^d) \times \mathbb{R},$$

with coordinates

$$(\rho, S, z).$$

The contact Hamiltonian has the schematic form

$$H^c(\rho, S, z) = K(\rho, S) + F(\rho) + B(\rho) + \gamma z.$$

Here K is kinetic energy, F contains the potential and diffusion-induced terms, B is an entropy contribution, and γz is the explicitly non-conservative part.

The contact Hamiltonian equations have the form

$$\partial_t \rho_t = -\nabla \cdot (\rho_t \nabla S_t),$$

$$\partial_t S_t = -\frac{1}{2}\|\nabla S_t\|^2 + \frac{1}{2}\varepsilon^2 \frac{\delta I}{\delta \rho}(\rho_t) + U - \gamma S_t - \varepsilon\gamma \log \rho_t,$$

and

$$\partial_t z_t = \int \left(\frac{1}{2}\|\nabla S_t(x)\|^2 + U(x)\right) \rho_t(x)\, dx + \frac{1}{2}\varepsilon^2 I(\rho_t) - \int \varepsilon\gamma(\log \rho_t(x) - 1)\rho_t(x)\, dx - \gamma z_t.$$

Remark 5.6. The terms involving γ are the signature of non-conservative dynamics. When $\gamma = 0$, the extra contact effects disappear and the system reduces to the conservative Hamiltonian bridge picture.

5.9 Contact Wasserstein Geodesics

The contact Hamiltonian flow lives on the extended phase space $T^*\mathcal{P}^+(\mathbb{R}^d) \times \mathbb{R}$. Its projection can be interpreted as a geodesic on an augmented probability manifold

$$\mathcal{P}^+(\mathbb{R}^d) \times \mathbb{R}.$$

The associated metric is again a Jacobi-type rescaling of the Wasserstein metric:

$$\widetilde{g}_J = \big(H^c - F(\rho) - B(\rho)\big) g_{W_2}.$$

This metric is still Wasserstein in its tangent directions, but the cost of moving through probability space is modulated by the evolving non-conservative energy.

Definition 5.7 (Contact Wasserstein geodesic)**.** A contact Wasserstein geodesic is a probability path, together with an accumulated action coordinate z_t, that solves the non-conservative bridge problem and is equivalently described as a geodesic for the contact Jacobi-type metric on $\mathcal{P}^+(\mathbb{R}^d) \times \mathbb{R}$.

5.10 Finite-Dimensional Parameterization

The full space $\mathcal{P}^+(\mathbb{R}^d)$ is infinite-dimensional. To compute bridges in practice, one restricts attention to a finite-dimensional family of pushforward distributions. Let λ be a reference distribution and let

$$T_{\theta_K} \circ \cdots \circ T_{\theta_1} \circ T_{\theta_0}$$

be a composition of maps. This induces a discrete sequence

$$\rho^\theta_{t_{k+1}} = (T_{\theta_{k+1}})_\# \rho^\theta_{t_k}.$$

This is exactly the geometric role played by a residual network: each block is a small transport step, and the whole network is a discretized path in probability space.

The training loss has three conceptual parts:

$$\ell = \underbrace{d^2_{W_2}(\rho^\theta_{t_K}, \rho_b)}_{\text{terminal marginal}} + \underbrace{\sum_{m=1}^{M} d^2_{W_2}(\rho^\theta_{t_{k_m}}, \rho_m)}_{\text{intermediate marginals}} + \underbrace{\sum_{k=1}^{K} \Phi_{t_k} d^2_{W_2}(\rho^\theta_{t_k}, \rho^\theta_{t_{k-1}})}_{\text{contact geodesic energy}}.$$

The scalar Φ_{t_k} is the discrete version of the contact Jacobi factor. It tells the model how expensive it is to move through that portion of the probability path.

Remark 5.8. This is the computational bridge between the geometry and generative modeling. The abstract object is a geodesic in an infinite-dimensional Wasserstein-type space. The implemented object is a sequence of parameterized pushforwards trained to match marginals while minimizing a geometry-weighted transport energy.

5.11 Guidance as Metric Modulation

Suppose we want the bridge to satisfy a condition

$$y = f(x_{t_s})$$

at some time t_s. A natural way to impose this condition is to add the penalty

$$\|y - f(x_{t_s})\|^2$$

to the action. Geometrically, this modifies the Jacobi factor:

$$\widetilde{g}'_J = \left(\Phi_t + \|y - f(x_{t_s})\|^2\right) g_{W_2}.$$

Regions of probability space that violate the condition become more expensive. The bridge is therefore steered without abandoning the underlying transport geometry.

5.12 Interpretation for Generative Policies

For generative policies, a policy is a pushforward

$$\pi_\theta(\cdot|s) = f_\theta(s, \cdot)_\# \rho_0.$$

Training changes the induced distribution over actions. The conservative Wasserstein-gradient-flow picture says that policy improvement is a mass-preserving transport process driven by a variational objective.

The contact-Wasserstein picture suggests a richer possibility:

$$\pi_\beta \longrightarrow \pi_Q$$

may be better modeled as a non-conservative bridge whose effective energy can change during learning.

In this interpretation:

- the behavior policy is the initial marginal,

- the improved soft policy is the terminal marginal,
- value information acts like a potential or guidance term,
- entropy controls exploration,
- trust-region effects can be encoded as transport costs,
- offline-to-online updates can inject or dissipate energy.

This is the conceptual reason for studying contact Wasserstein geometry in the context of drifting policies. It gives a language for distributional learning dynamics that are still geometric, but no longer forced to behave as closed conservative systems.